

Product Description

GFTP-510 **Fiberglass Reinforced** Thermal Gap Pad is a thermally conductive interface material engineered to provide efficient heat transfer between components and heat sinks, while accommodating surface irregularities and minimizing mechanical stress in low-pressure applications. Designed for assemblies requiring durable and stable interface materials, the pad incorporates fiberglass reinforcement to improve tear resistance, handling robustness, and dimensional stability, reducing the risk of material damage or deformation during installation and use. The material combines a ceramic-filled silicone formulation with a reinforced fiber structure to deliver consistent thermal and electrical performance, while maintaining controlled compressibility and long-term interface reliability across a range of operating conditions.

Key Features

- **Fiberglass-reinforced**
- Improved tear / puncture resistance
- Dimensional stability under compression
- Controlled deformation (less collapse / squeeze-out)
- Better handling during assembly (no tearing)
- Improved long-term interface stability (reduced creep / drift)

Applications

- Interfaces with sharp edges requiring cut-through resistance
- High clamping force or torque-loaded assemblies
- Server SSDs
- Manufacturing and rework environments requiring durability
- Large tolerance stack-ups needing dimensional stability
- Electrically isolating interfaces where mechanical damage must be avoided

Application Guidelines

- Avoid exposure to open flame and direct sunlight.
- Ensure application surfaces are clean.

Storage and Shelf Life

- Store at 8°C to 28°C in a dry environment (RH<70%).
- Shelf life is 12 months from the date of manufacture.

Packaging Information

- Available in thicknesses from 0.5 mm to 2.0 mm, with additional thicknesses available upon request
- Supplied in standard sheet formats and custom die-cut parts for specific application requirements
- Custom sizes and configurations supported to meet design and manufacturing needs

Property	Value	Test Method
Color	Green	Visual
Hardness (Shore 00)	50	ASTM D2240
Thermal Conductivity (W/(m·K))	5.0	ASTM D5470
Thickness (mm)	0.5-2.0	--
Density (g/cm ³)	3.3	ASTM D792
Breakdown Voltage (kV/mm)	>10	ASTM D149
Operating Temperature Range (°C)	-40 to 180	--
Volume Resistivity (Ω·cm)	10 ¹³	ASTM D257
Flammability Rating	V-0	UL94

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.